

Exploratory Data Analysis (EDA) Midterm Proposal

Prepared for Prof. Vishwantha Rao, HCC ITAI 1371 Machine Learning

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This midterm proposal was prepared for ITAI 1371 Machine Learning by group 6171 16347 3. Our team has reviewed the proposal for Prof. Rao's consideration. Our proposal was subsequently revised based on feedback from Prof. Rao.

GOAL 1: Prepare the Netflix Movies and TV Shows dataset for a future machine learning model that classifies whether a movie may be appropriate for a children's profile based on available movie characteristics.

GOAL 2: Focus the project on movies by filtering out television shows through reproducible Python code.

GOAL 3: Complete the required exploratory data analysis and preprocessing in a reproducible Jupyter/Colab notebook without manually changing the original dataset.

GOAL 4: Demonstrate our understanding of the course material by completing the following requirements:

1. Load and inspect the unchanged raw dataset in Python using pandas.
2. Create a stratified 70% training and 30% testing split in Python before any preprocessing values are learned, and fit every imputation, transformation, encoding, and balancing step on the training set only.
3. Identify and handle NaN and null values using appropriate, documented methods.
4. Scale numerical features and normalize skewed distributions when justified by the data.
5. Encode useful categorical features, including one-hot encoding where appropriate.
6. Examine target-class balance and apply a balancing method only if meaningful imbalance is found.
7. Engineer features that support the future classification problem.
8. Use Matplotlib to show meaningful before-and-after distributions and print sample records during processing.
9. Document the complete pipeline so the cleaned data can be reused for the final ML project.

PROPOSAL

This project will investigate whether machine learning can classify a Netflix movie as potentially appropriate for a children's profile using characteristics such as genre, release year, duration, country, when the movie was added to Netflix, the number of cast members and directors, and description length. The project will prepare and analyze the data through a reproducible Python notebook. The cleaned dataset will later support the development and evaluation of classification models for the final project.

METHODOLOGY

Use the Netflix Movies and TV Shows dataset <https://www.kaggle.com/datasets/shivamb/netflix-shows>.

The raw dataset contains 8,807 rows and 12 columns, exceeding the assignment's 2,000-row minimum. Each row represents a Netflix title, with fields including type, title, director, cast, country, date added, release year, rating, duration, listed genres, and description.

1. Load the unchanged Kaggle file into Colab using Python code. Inspect its dimensions, column types, missing values, and rating labels without manually editing the source file. A small number of records have a run time typed into the rating field; the notebook moves that value to the duration field in code.
2. Use Python to filter the intended project population to movies (type = Movie). Because every retained record has the same type value, the type field is not used as a model input.
3. Create the binary kid_safe target from the rating field. Ratings G, PG, TV-G, TV-PG, TV-Y, TV-Y7, and TV-Y7-FV are coded as 1 (kid-safe), and PG-13, R, NC-17, TV-14, and TV-MA are coded as 0 (mature). Movies rated NR or UR, or with no rating, have no known maturity label, so they are counted and excluded from the modeling table rather than automatically labeled unsafe.
4. Use the original rating field only to construct the target, then exclude it from the model inputs. Using rating as both the source of the target and a predictor would create target leakage.
5. Split the labeled movie population programmatically into a stratified 70% training set and 30% testing set using a fixed random seed. The raw data is inspected before the split, but every value learned from the data (medians, skew corrections, scaling ranges, category lists, and the balancing decision) is determined from the training set only. Keep the test data untouched, with its natural class mix, for future evaluation.
6. Handle missing values with documented methods learned from the training set: numeric gaps (such as duration or date added) are filled with the training median, and a missing country becomes an explicit Unknown category. Because many movies lack director, cast, or country information, the notebook also records missing-value flags and counts of listed people, so that missingness is kept as information. Document each decision in the notebook.
7. Engineer features from the available movie information: duration in minutes, release year, year and month added to Netflix, years between release and arrival on Netflix, number of cast members, directors, countries, and genres, description word count, title length, three missing-value flags (director, cast, country), the first listed country, and the list of genres. Title, show ID, and the raw director, cast, and description text are not used as model inputs.
8. Encode and transform the features: one-hot encode the first listed country (the ten most common countries plus Other), multi-hot encode the genre list because a movie can have several genres, and ordinal-encode a duration bucket (Short, Standard, Long). Normalize strongly skewed numerical features (absolute skew above 0.75) with a Yeo-Johnson power transform, then scale all numerical features to a 0 to

1 range with min-max scaling. Unusually long or short movies are real data and are kept, not removed as outliers. All encoders and transformers are fitted on the training set and then applied to the test set.

9. Use Matplotlib to compare meaningful distributions before and after preprocessing, including missing values before and after filling, skewed numerical features before and after normalization and scaling, and the target distribution before and after balancing. Print sample rows after major stages so the notebook clearly demonstrates how the data changes.

10. Examine the `kid_safe` target distribution in the training set. If the minority-to-majority class ratio is below 0.8, balance the training data by randomly oversampling the minority class. The test data is never balanced, so it continues to represent unseen data.

11. Export the cleaned, model-ready datasets (balanced training set, unbalanced training set, and untouched test set, each containing the encoded features and the `kid_safe` target) and upload the required materials to GitHub by 03-Oct-2026: the original dataset URL document, proposal PDF, Jupyter notebook, cleaned datasets, and a separate reflection/contribution journal containing an entry from every team member.

EXPECTED RESULTS

We expect the movie-only dataset to remain above the assignment's 2,000-row minimum after television shows are removed and unrated movies are set aside through Python. The training-data analysis will likely identify missing values in fields such as director, cast, country, date added, rating, and duration. Missing and unrated observations will be described accurately rather than treated as statistical outliers.

We also expect the engineered `kid_safe` target to be imbalanced because kid-safe movies may be less common than mature-rated movies. The notebook will measure the actual class distribution first and will balance only if the minority-to-majority ratio is below 0.8, using random oversampling of the minority class on the training data only, so the test set continues to represent unseen data.

The completed notebook will document every preprocessing decision and display sample records before and after major processing stages. Matplotlib visualizations will compare meaningful distributions, such as missing-value patterns, numerical-feature distributions before and after normalization and scaling, and the target distribution before and after class balancing. All transformations will be completed through Python without manually changing the original Kaggle file.

The final output will be cleaned, reproducible datasets with appropriate missing-value handling, encoded categorical variables (one-hot country, multi-hot genre, and an ordinal duration bucket), normalized and scaled numerical features, and engineered features derived from the available movie information. The original rating column will not be included among the model inputs because it is used to construct the prediction target.

This prepared dataset will support the group's final binary-classification project. The untouched 30% test set will be retained for future model evaluation, and each group member's work will be documented separately in the required reflection and contribution journals.